

Note 8.1 Integration by Parts

Product Rule in Integral Form

- **Definition of the integration by parts**

Integration by Part Formula

$$\int f(x)g'(x)dx = f(x)g(x) - \int f'(x)g(x) dx$$

or,

$$\int u dv = uv - \int u dv$$

Proof:

By the [product rule](#),

$$\frac{d}{dx}(f(x)g(x)) = \frac{d}{dx}f(x) \cdot g(x) + \frac{d}{dx}g(x) \cdot f(x)$$

integrate both side of the equation,

$$f(x)g(x) = \int f'(x)g(x) dx + \int f(x)g'(x) dx$$

which is equivalent to the equation.

- **Reduction Formula:**

The replacement of a formula that is in a more simple representation.

- Example:

Obtain a formula that expresses the integral

$$\int \cos^n x dx$$

in terms of an integral of a lower power of $\cos x$.

$$\begin{aligned}\int \cos^n x dx &= \cos^{n-1} x \sin x + (n-1) \int \sin^2 x \cos^{n-2} x dx \\ &= \cos^{n-1} x \sin x + (n-1) \int (1 - \cos^2 x) \cos^{n-2} x dx \\ &= \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx - (n-1) \int \cos^n x dx\end{aligned}$$

and if we move a term,

$$n \int \cos^n x dx = \cos^{n-1} x \sin x + (n-1) \int \cos^{n-2} x dx$$

then,

$$\int \cos^n x \, dx = \frac{\cos^{n-1} x \sin x}{n} + \frac{n-1}{n} \int \cos^{n-2} x \, dx$$

Evaluating Definite Integrals by Parts

- **Definition:**

Integration by parts formula for definite integrals

$$\int_a^b f(x)g'(x) \, dx = f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) \, dx$$

Proof: By the [fundamental theorem](#),

$$\begin{aligned} \int_a^b f(x)g'(x) \, dx &= \int f(x)g'(x) \, dx \Big|_a^b \\ &= \left[f(x)g(x) - \int f'(x)g(x) \, dx \right]_a^b \\ &= f(x)g(x) \Big|_a^b - \int_a^b f'(x)g(x) \, dx \end{aligned}$$

Exercises

- P460 35,47